Hoàng Kim Vĩnh

Embedded Firmware Engineer



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Education

Information Technology (Vietnamese-Japanese)

Hanoi University of Science and Technology — 2023 – 2027

Final-year Information Technology (Vietnamese-Japanese) student, taught in Japanese. My focus is embedded systems and IoT, built around microcontrollers, real-time systems and computer networks; the machine learning coursework feeds the models I run on-device. I lead most of the team projects I take part in.

Skills

Embedded GUI

TouchGFX on STM32, LVGL on ESP32, GUI design tooling, `MIPI-DSI display interface`, I/O expander for touch and buttons, Layout and screen-state design

Programming Languages

C (C99, Embedded C), C++, Python, basic Assembly

Firmware & RTOS

STM32 Cortex-M series, STM32Cube HAL, Bare-metal register work, Hierarchical state machines, Non-blocking + soft timers, FreeRTOS, ESP-IDF, PlatformIO, Bootloader / IAP, USB Host, FATFS

Protocols

`CAN 2.0B 29-bit`, `FDCAN`, `SPI-to-CAN bridge`, `OCPP`, `GB/T 27930`, `DLT-645`, `MQTT - topic design, QoS, retain, LWT`, `UART / RS485 / Modbus`, SPI Ethernet controller, Wi-Fi provisioning, Framed ASCII protocol with checksum + resync

Hardware & power

EasyEDA Pro, Contactor, power module, current/voltage ramping, Insulation monitoring (IMD), Safe sequential switching, Logic analyser, Fault injection, Shift register for LED bus, RTD temperature sensing

Edge AI & App

Computer vision, PyTorch, On-device inference, OpenCV, Python, FastAPI, TypeScript, NestJS, PostgreSQL, Flutter, React

Languages

Japanese (JLPT N3)

Vietnamese (Native)

English (Working proficiency)

Profile

I am a final-year Information Technology (Vietnamese-Japanese) student at Hanoi University of Science and Technology, with a focus on embedded systems and IoT. Most of my firmware work happens at APES Lab, where I contribute to a commercial EV charging station developed with a partner manufacturer. The code is under NDA, but the technical scope is what I do every day.

My AI background is formal rather than self-taught (DeepLearning.AI × Stanford), and it is what lets me run models on-device: a computer vision pipeline for vehicle damage assessment, and a Vietnamese voice-command classifier that runs entirely on an ESP32-P4 with no cloud dependency.

Experience

Present — **APES Lab** / **Embedded Firmware Engineer - EV Charging Station**

Contribute to a commercial EV charging station developed with a partner manufacturer. The code is under NDA, so the scope below describes my responsibilities rather than the implementation.

- Power-sharing firmware for a ring-topology EV charging network - STM32H723, CAN 2.0B 29-bit over SPI-to-CAN MCP2515, up to 32 charging guns across 16 cabinets.
- Distributed power borrowing and reclaim algorithms - the firmware autonomously borrows capacity from idle guns and reclaims it later.
- Four-layer architecture: Application -> Middleware (Service / FSM / Control) -> Driver -> HAL, with strictly enforced upward-only calls.
- MISRA C:2012 static analysis pipeline (cppcheck with a custom addon) and PC-based unit tests with a mocked HAL.
- Non-blocking firmware design - no HAL_Delay() anywhere; all timing comes from a 1 ms software timer array.
- Two-gun EV charger firmware - OCPP, GB/T 27930 vehicle-side comms, DLT-645 DC metering, WIZnet Ethernet controllers, RFID, insulation monitoring, USB Host with FATFS, bootloader / IAP.

Summer Internship — **Yootek** / **IoT Firmware Engineer**

SmartGarden - an enterprise IoT platform.

- ESP32 firmware on ESP-IDF, publishing sensor state over MQTT.
- NestJS backend with PostgreSQL and Prisma, WebSocket for live updates.
- JWT authentication and role-based access control.

2023 – 2027 — **Academic Projects** / **Embedded & IoT Engineer - Team Lead**

Led or co-led most team projects from first year to the present; personal projects were delivered solo, end to end across hardware, firmware and app. Technical detail for each project is in the Projects section below.

Projects

`embedded-iot-toolkit` Open source - MIT - 89 tests

A dependency-free C99 library of the firmware building blocks I keep rewriting: a non-blocking timer bank, an ISR-safe framed serial protocol with checksums and resynchronisation, a CAN 2.0B 29-bit codec with a bit-timing solver, and sensor decoders that report staleness instead of returning a stale value.

`github.com/CatKod/embedded-iot-toolkit`

`rokid-max-diy-station` Open source - MIT - 218 tests

Rokid Max AR glasses turned into a spatial display: a virtual Windows monitor anchored in the room, not to your eyes. A Mahony filter fuses the USB HID IMU into a 3-DoF head pose, an IddCx driver creates a real monitor so the OS owns the input, and D3D11 anchors that desktop in 3D.

`github.com/CatKod/rokid-max-diy-station`

`SmartHome` University capstone - 10/10

A two-tier architecture where the real-time MCU never depends on the cloud for safety. The ESP32-S3 handles Wi-Fi and MQTT; the STM32H7 owns the hardware. RFID and PIN entry, a shift-register LED bus, environmental sensors and an automatic window actuator.

`github.com/CatKod/SmartHome`

`Smart Power Outlet` Graduation capstone - solo

A complete IoT product rather than a board: the PCB was designed in EasyEDA Pro, the firmware is ESP-IDF with Wi-Fi provisioning so no credential is ever hardcoded, a PlatformIO build covers both ESP32 and ESP8266, and a cross-platform Flutter app drives it.

`github.com/CatKod/Smart_Power_Outlet`

`Sun-Car Damage Assessment` SUN scholarship project

A computer vision pipeline for object detection, instance segmentation and severity scoring, built to apply for a SUN scholarship. An ESP32-CAM captures frames and streams the result to an STM32 over UART, so inference sits on the edge rather than in the cloud.

`github.com/CatKod/Sun-Car-Damage-AI-Assessment`

Certifications

Japanese Language Proficiency Test - N3

Machine Learning Specialization

`DeepLearning.AI x Stanford University · Apr – May 2025`

- Supervised ML: Regression & Classification
- Advanced Learning Algorithms
- Unsupervised Learning, Recommender Systems & RL
- ML Capstone

Deep Learning Specialization

`DeepLearning.AI · Jul 2025`

- Neural Networks and Deep Learning